32. Construct a C Program to Simulate the Least Recently Used (LRU) Paging Technique of Memory Management

Code - Part 1 (Lines 1–35)

```c
1  #include <stdio.h>
2
3  #define MAX_FRAMES 3
4
5  // Print frames
6  void printFrames(int frames[], int n) {
7      for (int i = 0; i < n; i++) {
8          if (frames[i] == -1)
9              printf(" - ");
10         else
11             printf(" %d ", frames[i]);
12     }
13     printf("\n");
14 }
15
16 int main() {
17
18     int frames[MAX_FRAMES];
19     int usageHistory[MAX_FRAMES];
20
21     // Initialize
22     for (int i = 0; i < MAX_FRAMES; i++) {
23         frames[i] = -1;
24         usageHistory[i] = 0;
25     }
26
27     int pageFaults = 0;
28
29     int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2};
30     int n = sizeof(referenceString) / sizeof(referenceString[0]);
31
32     printf("Reference String: ");
33     for (int i = 0; i < n; i++) {
34         printf("%d ", referenceString[i]);
35     }
```

Code - Part 2 (Lines 36–68)

```
36
37      printf("\n\nPage Replacement Order:\n");
38
39      for (int i = 0; i < n; i++) {
40
41          int page = referenceString[i];
42          int pageFound = 0;
43
44          // Check HIT
45          for (int j = 0; j < MAX_FRAMES; j++) {
46              if (frames[j] == page) {
47                  pageFound = 1;
48
49                  // update usage
50                  for (int k = 0; k < MAX_FRAMES; k++) {
51                      if (k != j)
52                          usageHistory[k]++;
53                  }
54                  usageHistory[j] = 0;
55                  break;
56              }
57          }
58
59          // MISS → Replace LRU
60          if (!pageFound) {
61
62              int lruIndex = 0;
63
64              for (int j = 1; j < MAX_FRAMES; j++) {
65                  if (usageHistory[j] > usageHistory[lruIndex]) {
66                      lruIndex = j;
67                  }
68              }
69
```

Code - Part 3 (Lines 69–95)

```
69
70              int replacedPage = frames[lruIndex];
71              frames[lruIndex] = page;
72              usageHistory[lruIndex] = 0;
73
74              for (int k = 0; k < MAX_FRAMES; k++) {
75                  if (k != lruIndex)
76                      usageHistory[k]++;
77              }
78
79              printf("Page %d -> ", page);
80
81              if (replacedPage == -1)
82                  printf("Load into empty frame: ");
83              else
84                  printf("Replace %d with %d: ", replacedPage, page);
85
86              printFrames(frames, MAX_FRAMES);
87
88              pageFaults++;
89          }
90      }
91
92      printf("\nTotal Page Faults: %d\n", pageFaults);
93
94      return 0;
95  }
```

Output

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2

Page Replacement Order:
Page 7 -> Load into empty frame:  7  -  -
Page 0 -> Load into empty frame:  7  0  -
Page 1 -> Load into empty frame:  7  0  1
Page 2 -> Replace 7 with 2:  2  0  1
Page 3 -> Replace 1 with 3:  2  0  3
Page 4 -> Replace 2 with 4:  4  0  3
Page 2 -> Replace 3 with 2:  4  0  2
Page 3 -> Replace 0 with 3:  4  3  2
Page 0 -> Replace 4 with 0:  0  3  2

Total Page Faults: 9


...Program finished with exit code 0
Press ENTER to exit console.
```